

VALIDITY OF CRITICAL SPEED OF CRITICAL SPEED DETERMINED FROM MULTIPLE TIME TRIALS IN MIDDLE- AND LONG-DISTANCE RUNNERS.

Kieron Lester¹, Stuart Warmington¹, Rodney Siegel¹, and Samantha Hoffmann¹

¹ Centre for Sport Research in the Institute for Physical Activity and Nutrition, Deakin University

Email: kdles@deakin.edu.au

INTRODUCTION

Physiological profiling provides coaches and sport scientists with objective physiological data to better gauge athletic performance potential, quantify and evaluate training adaption, and inform decisions related to training prescription [1-4].

The National Testing Protocol for the physiological profiling of middle-distance (MD) and long-distance (LD) runners in Australia includes an incremental step test and a maximal accumulated oxygen deficit (MAOD) test, and provide information related to aerobic and anaerobic performance determinants [4]. Previous findings from our research group (unpublished) revealed that only 25% of coaches and sport scientists working with nationally categorised MD and LD runners in Australia are using the entire National Testing Protocol as it was intended. When asked what could be improved 81% provided a response related to outcome measures, with 25% specifically interested in critical speed (CS). Practitioners also expressed a preference for field-based tests over lab-based protocols, and when asked about their perceptions on CS testing protocols, time trials were the only method to reach consensus ($\geq 80\%$ agreement).

This study aims to: (1) determine if estimates of CS from three time trials are valid, and (2) understand if there are any trends in validity across race distance preferences.

METHODS

Participants will attend five visits to a 400-m track, with 24–48 hours separating each visit. Prior to each visit, participants will record their dietary and fluid intake, avoid alcohol and strenuous exercise for 24 hours and caffeine for 3 hours, and complete a wellness questionnaire assessing fatigue, soreness, stress, sleep quality and mood. Female participants will also complete a menstrual cycle diary and questionnaire, allowing trials to be scheduled within the early follicular phase to minimise the influence of ovarian hormones on physiological responses.

The initial three visits will consist of a 1200-m, 2400-m and 3600-m time trial, each completed in a random order. Each participant's CS will be calculated using the two-parameter hyperbolic model (CS2-hyp), the linear distance-time model (CSlinear) and the speed versus

1/time model (CP1/time), with the model producing the highest R^2 value selected as the representative CS value for each participant. Following completion of all three TT, participants will complete two constant-speed trials on the same track, each lasting 30 minutes or until volitional exhaustion. The first trial will be performed at 100% of each participant's CS. Based on the outcome of this trial, the pace for the second trial will be adjusted by $\pm 2.5\%$ to reflect the typical error associated with CS estimation: if the participant completed the full 30 minutes, pace will be increased by 2.5%; if they were unable to complete the trial, pace will be decreased by 2.5%.

Oxygen consumption, heart rate and blood lactate were recorded throughout each fixed-speed trial to determine whether the CS estimate derived from the TT method reflected the boundary between heavy and severe exercise intensity, indicated by a sustained rise in $\dot{V}O_2$ (slow component), heart rate exceeding 90% of heart rate reserve, and/or failure to complete the full 30 minutes.

CONCLUSIONS

In summary, this study aims to evaluate the validity of CS determined from three time trials against first-principle physiological variables in middle- and long-distance runners. The outcomes of this study will establish whether the TT method provides an accurate reflection of the boundary between heavy and severe exercise intensity. Should this method prove valid, it will offer coaches, sport scientists and athletes a practical and time-efficient alternative to the current National Testing Protocol, addressing the implementation barriers previously identified by coaches and sport scientists. This is significant because it will allow athlete strengths and weaknesses to be better quantified, training to be more individually and specifically prescribed, and ultimately support more informed selection and performance decisions for middle- and long-distance runners at the highest level.

REFERENCES

- [1] Faude O et al. *Sports Med* **39**: 469-490, 2009.
- [2] Galan-Rioja et al. *Sports Med* **50**: 1771-1783, 2020.
- [3] Seiler S et al. *Sports Sci* **13**: 1-27, 2009.
- [4] Tanner R et al. *Human Kinetics*, 2012.

1. Faude, O., W. Kindermann, and T. Meyer, *Lactate threshold concepts*. Sports medicine, 2009. **39**(6): p. 469-490.
2. Galan-Rioja, M.A., et al., *Relative Proximity of Critical Power and Metabolic/Ventilatory Thresholds: Systematic Review and Meta-Analysis*. Sports Med, 2020. **50**(10): p. 1771-1783.
3. Seiler, S. and E. Tønnessen, *Intervals, thresholds, and long slow distance: the role of intensity and duration in endurance training*. Sportscience, 2009. **13**.
4. Tanner, R. and C. Gore, *Physiological tests for elite athletes*. 2012: Human kinetics.